# Shenghan **Gao**

Photonic Integrated Circuits · Photonics Hardware · Validation & Test

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## Summary

Ph.D. candidate in Electrical and Computer Engineering specializing in integrated photonics, photonic hardware validation, and coherent electro-optic systems. End-to-end experience across silicon photonic integrated circuit design, passive and active photonic devices, cleanroom fabrication, fiber-to-chip integration, RF/EO characterization, DSP-based signal analysis, and Python/PyVISA/SCPI automated measurement. Seeking PIC design, silicon photonics, photonics validation/test, electro-optic device, and optical hardware R&D roles.

## Core Skills

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| **PIC design** | Silicon photonics and TFLN/LNOI PIC design; waveguides, directional couplers, interferometers, electro-optic modulators, thermal phase shifters |
| **Simulation/layout** | GDSFactory, Cadence Virtuoso, KLayout; Lumerical, RSoft, COMSOL, Zemax/CodeV; optical mode analysis, device optimization, PIC layout preparation |
| **Fab/integration** | Cleanroom fabrication, spin coating, photoresist processing, electron-beam lithography, etching, SEM inspection; fiber-array coupling, fiber-to-chip coupling, photonic packaging, optical alignment |
| **Validation** | Device characterization, high-speed EO/RF characterization, VNA, RF detection, high-speed/balanced photodetection, EOMs, OSAs, SVAs, spectrum analyzers, oscilloscopes, tunable lasers |
| **Automation/DSP** | Python measurement automation, PyVISA/SCPI, NI MAX, automated testing and data acquisition, parameter sweeps, data logging, FFT/frequency-domain analysis, DSP equalization, MATLAB |
| **Coherent optics** | Optical injection locking, semiconductor-laser rate equations, coherent interference, carrier recovery, AM/PM modulation transfer, PID phase stabilization |

## Education

**University of Pittsburgh** Ph.D. in Electrical and Computer Engineering — *Dec. 2026 (expected)*

**Georgia Institute of Technology** M.S. in Computer Science — *Dec. 2026 (expected)*

**University of Rochester** M.S. in Optics — *Aug. 2019*

**University of Rochester** B.S. in Optical Engineering | B.S. in Applied Mathematics — *Aug. 2017*

## Research and Industry Experience

### University of Pittsburgh — *Pittsburgh, PA*

Research Assistant | Advisor: Nathan Youngblood — *Spring 2020 - Present*

Developed a distributed photonic computing testbed that uses optical injection locking to recover a phase-coherent carrier at a remote node.

- Mapped injection-ratio and detuning ranges for stable carrier recovery; compared fiber-link measurements against calibrated laser models.
- Designed PIC layouts with waveguides, couplers, and interferometers for chip-scale integration; supported e-beam fabrication, SEM inspection, and fiber-to-chip alignment.
- Built a balanced-detection testbed with PID phase feedback; automated instrument sweeps in Python/PyVISA and characterized modulation transfer with FFT/DSP.

### Lumentum Operations LLC — *San Jose, CA*

Research Intern | Mentor: Joe Wen — *May 2023 - Aug 2023*

Developed receiver-side DSP to recover communication waveforms distorted by intersymbol interference.

- Implemented FIR equalizers in MATLAB and Python; tuned filter coefficients using FFT analysis of measured signals.
- Compared filter responses on measured waveforms and validated demodulated-signal recovery across test cases.

### University of Rochester — *Rochester, NY*

Research Assistant | Advisor: Xi-Cheng Zhang — *Sep 2017 - Aug 2019*

Investigated liquid plasmas as terahertz sources by exciting liquids with ultrashort laser pulses.

- Generated THz signals and characterized how liquid temperature, polarity, and viscosity affected emission strength, bandwidth, and spectral content.
- Analyzed time- and frequency-domain measurements to assess signal-to-noise ratio, sensitivity, and repeatability.

### Sunny Optics — *Ningbo, China*

Research Intern | Mentor: Chao Xu — *May 2018 - Aug 2018*

Designed optical systems for automotive LiDAR and compact mobile cameras.

- Optimized lens designs in CodeV/Zemax against optical performance, tolerance, and packaging requirements.
- Worked with opto-mechanical engineers during lab integration, using alignment and test feedback to refine the optical systems.

## Selected Publications


[1] **S. Gao**, K. Lüdge, F. Da Ros, and N. Youngblood, "Distributed coherent optical computing via injection-locked photonic networks," submitted to *Applied Physics Letters*, 2026 (under review). arXiv:2604.13450.

[2] **S. Gao**, P. R. Swinehart, S. Zhong, J. Wuenschell, N. Lalam, M. Buric, and K. P. Chen, "Strain-free FBG sensors for accurate temperature measurements from cryogenic temperatures to 300 K," *Proc. SPIE*, vol. 12532, 125320U, 2023. SPIE Defense + Commercial Sensing (oral presentation). doi:10.1117/12.2670079.

[3] Q. Jin, Y. E., **S. Gao**, and X.-C. Zhang, "Preference of subpicosecond laser pulses for terahertz wave generation from liquids," *Advanced Photonics*, vol. 2, no. 1, 015001, 2020. doi:10.1117/1.AP.2.1.015001.

[4] **S. Gao**, Q. Jin, Y. E., and X.-C. Zhang, "Investigation of liquid properties on emitting terahertz wave under ultrashort optical excitation," *Proc. SPIE*, vol. 11196, 1119609, 2019. doi:10.1117/12.2538433.


## Selected Talks

- **S. Gao**, "Distributed coherent photonic computing via injection locking," SPIE Optics + Photonics (oral presentation), 2026.
- **S. Gao**, "Distributed coherent optical computing via injection-locked photonic networks," Pittsburgh Quantum Institute (invited Ph.D. candidate talk), 2025.